NAME: MANOJ A C

SRN: PES1UG25CS826, H SECTION

NAME: RUTHIK

SRN: PES1UG25CS837, H SECTION

SCREENSHOT: COMMANDS FOR RUNNING ALL TASKS

```
seed@seed:~$ cd ~/OS-Jackfruit/boilerplate

# Stop any old containers
sudo ./engine stop alpha beta cpu-high cpu-low 2>/dev/null

# Clear logs
sudo rm -rf logs
mkdir -p logs

# Start containers
echo "=== Starting containers ===="
sudo ./engine start alpha ../rootfs-alpha "sh -c 'echo Alpha message'" --soft-mib 50 --hard-mib 100
sudo ./engine start beta ../rootfs-beta "sh -c 'echo Beta message'" --soft-mib 50 --hard-mib 100

sleep 2

# Show containers
echo ""
echo "=== Container list ===="
sudo ./engine ps

# Show logs
echo ""
echo "=== Alpha Log ===="
cat logs/alpha.log

echo ""
echo "=== Beta Log ===="
cat logs/beta.log

# CPU Scheduling Test
echo ""
echo "=== CPU Scheduling Test ===="
sudo cp cpu_hog ../rootfs-alpha/ 2>/dev/null
sudo cp cpu_hog ../rootfs-beta/ 2>/dev/null

sudo ./engine start cpu-high ../rootfs-alpha "./cpu_hog" --nice -20
sudo ./engine start cpu-low ../rootfs-beta "./cpu_hog" --nice 19

sleep 5
```

SCREENSHOT: SHOWING

- **MULTIPLE CONTAINERS**
- **LOG MESSAGES**
- **CPU SCHEDULING TEST**
- **CPU USAGE**
- **CLEAN UP**
- **ZOMBIE PROCESS**

```
echo ""
echo "=== CPU Usage (high priority should show higher %) ==="
sudo dmesg | grep container_monitor | tail -10No zombies found - GOOD!"
[sudo] password for seed:
Container alpha not found
=== Starting containers ===
Container alpha started with PID 3957
Container beta started with PID 3962

=== Container list ===
ID          PID     STATE     SOFT(MB)     HARD(MB)
beta        3962    running   50           100
alpha       3957    running   50           100


Brave Web Browser

=== Beta Log ===
Beta message

=== CPU Scheduling Test ===
Container cpu-high started with PID 3979
Container cpu-low started with PID 3984

=== CPU Usage (high priority should show higher %) ===
    PID  NI COMMAND         %CPU
   3979 -20 cpu_hog          100
   3984  19 cpu_hog          100

=== Clean up ===
Sent SIGTERM to container alpha (PID 3957)
Sent SIGTERM to container beta (PID 3962)
Sent SIGTERM to container cpu-high (PID 3979)
Sent SIGTERM to container cpu-low (PID 3984)

=== Checking for zombie processes ===
No zombies found - GOOD!
```

SCREENSHOT: KERNAL MODULE MESSAGES

```
=== Kernel module messages ===
[  285.881214] [container_monitor] Module loaded. Device: /dev/container_monitor
[  313.647720] [container_monitor] Registering container=alpha pid=3957 soft=52428800 hard=104857600
[  313.666523] [container_monitor] Registering container=beta pid=3962 soft=52428800 hard=104857600
[  314.549365] [container_monitor] Process pid=3957 (container=alpha) exited
[  314.549427] [container_monitor] Process pid=3962 (container=beta) exited
[  315.729851] [container_monitor] Registering container=cpu-high pid=3979 soft=41943040 hard=67108864
[  315.741555] [container_monitor] Registering container=cpu-low pid=3984 soft=41943040 hard=67108864
[  316.597079] [container_monitor] Process pid=3979 (container=cpu-high) exited
[  316.597084] [container_monitor] Process pid=3984 (container=cpu-low) exited
seed@seed:~/OS-Jackfruit/boilerplate$
```

SCREENSHOT: SUPERVISOR RUNNING

```
gcc -O2 -Wall -static -o memory_hog memory_hog.c
gcc -O2 -Wall -static -o cpu_hog cpu_hog.c
gcc -O2 -Wall -static -o io_pulse io_pulse.c
Ubuntu-Server 22.04.5 LTS amd64
-176-generic/build M=/home/seed/OS-Jackfruit/boilerplate modules
make[1]: Entering directory '/usr/src/linux-headers-5.15.0-176-generic'
  CC [M]  /home/seed/OS-Jackfruit/boilerplate/monitor.o
  MODPOST /home/seed/OS-Jackfruit/boilerplate/Module.symvers
  CC [M]  /home/seed/OS-Jackfruit/boilerplate/monitor.mod.o
  LD [M]  /home/seed/OS-Jackfruit/boilerplate/monitor.ko
  BTF [M] /home/seed/OS-Jackfruit/boilerplate/monitor.ko
Skipping BTF generation for /home/seed/OS-Jackfruit/boilerplate/monitor.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-176-generic'
Supervisor running. Control socket: /tmp/mini_runtime.sock
[LOG] Wrote 14 bytes to logs/alpha.log
Alpha message
[LOG] Wrote 13 bytes to logs/beta.log
Beta message
```